Sabarivishnu Balakrishnan

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Summary

Computer & Communication Engineering undergraduate specializing in cloud-native full-stack development (MERN), DevOps automation (Docker, Kubernetes, Terraform, CI/CD), and applied AI/RAG systems. AWS Certified Cloud Practitioner with three end-to-end personal projects spanning backend architecture, infrastructure automation, and machine learning.

Education

Amrita Vishwa Vidyapeetham, Coimbatore Aug 2023 – Jul 2027
B.Tech in Computer and Communication Engineering
- **CGPA: 7.59**
- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Design and Analysis of Algorithms

Technical Skills

Languages: Python, JavaScript, C, C++, SQL
Frontend: React.js, HTML5, CSS3, Tailwind CSS
Backend: Node.js, Express.js, FastAPI, Flask, RESTful APIs
Databases: MongoDB Atlas, MySQL, Redis, SQLite, ChromaDB
Cloud & DevOps: AWS, Docker, Kubernetes (K3s), Terraform, GitHub Actions, Prometheus, Grafana
AI/ML: LangChain, Google Gemini API, Prompt Engineering, TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, SciPy, Matplotlib
Tools: Git, GitHub, Linux, Postman

Projects

MedConnect — Cloud-Native Healthcare Platform | GitHub
- *React, Node.js, Express, MongoDB Atlas, Redis, Docker, Kubernetes, AWS EC2, Terraform, GitHub Actions, Prometheus, Grafana*
- Built a full-stack healthcare platform on the MERN stack with JWT authentication, role-based access control across admin/doctor/patient roles, and RESTful APIs for appointment scheduling.
- Added Redis caching, API rate limiting, request validation, and centralized logging to harden the backend for concurrent, multi-role usage.
- Containerized all services with Docker and deployed via Kubernetes (K3s) on AWS EC2 with MongoDB Atlas as the managed database, provisioning infrastructure as code with Terraform.
- Automated build-test-deploy pipelines with GitHub Actions and set up Prometheus/Grafana dashboards for live monitoring of application and infrastructure metrics.

AI DevOps Copilot — RAG-Based Log Analysis Assistant | GitHub
- *Python, FastAPI, React, LangChain, ChromaDB, Google Gemini API, Sentence Transformers, SQLite, Docker*
- Built a full-stack AI assistant that ingests infrastructure logs and returns automated diagnostics, replacing manual log-grepping across five platforms (Kubernetes, Docker, AWS CloudWatch, Jenkins, GitHub Actions).
- Implemented a Retrieval-Augmented Generation (RAG) pipeline with LangChain, ChromaDB, and Sentence Transformers to ground responses in a custom DevOps knowledge base instead of relying on the LLM alone.
- Integrated the Google Gemini API to generate structured output — root cause, recommended fix, verification command, and confidence score — for each analyzed log entry.
- Exposed the pipeline through FastAPI REST endpoints for log ingestion, AI analysis, and incident tracking, with persistent storage in SQLite.

Photovoltaic Fault Diagnosis using Deep Learning (Ongoing) | GitHub
- *Python, TensorFlow/PyTorch, NumPy, Pandas, Scikit-learn, SciPy, Matplotlib*
- Developing a CNN-based multi-task model to separate environmental PV faults (soiling, shading) from electrical faults (cracks, discoloration) using I–V curve analysis on real-world panel data.
- Engineered a multi-channel feature set — Residual Current, Current Ratio, and dI/dV slope — from raw I–V curves to improve fault discrimination beyond single-curve baselines.
- Building a validation framework grounded in Voc, Isc, and Fill Factor deviation analysis, with full preprocessing, EDA, and feature engineering pipelines.

Certifications

- AWS Certified Cloud Practitioner (CLF-C02)
- Meta Back-End Developer Professional Certificate